Predicting Squirrel–Human Interaction Behaviour Using Behavioural, Spatial, and Temporal Features

Group W14G02

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Executive Summary

This report studies whether squirrel behaviour toward humans can be predicted using data from the 2018 Central Park Squirrel Census dataset. The project focused on three types of behaviour: squirrels approaching humans, avoiding humans, and behaving neutrally. To answer this research question, several data preprocessing methods were used, including feature engineering, handling missing values, combining environmental data, and grouping location data into regions. Correlation analysis, machine learning models, and clustering techniques were then applied to explore patterns in squirrel behaviour.

The correlation analysis showed that no single feature had a strong relationship with squirrel and human interaction behaviour, which could mean that squirrel behaviour is influenced by multiple factors rather than a single one. Logistic Regression showed some ability to detect minority interaction behaviours but couldn't clearly separate the interaction categories. On the other hand, Random Forest performed better, achieving an accuracy of 73.6% and a weighted F1-score of 66.7%. The results also showed that spatial location was one of the most important factors affecting squirrel and human interaction behaviour. Also, clustering analysis identified meaningful behavioural and environmental groups linked to different interaction patterns.

Overall, the findings suggest that squirrel and human interaction behaviour is complex and influenced by behaviour, environment, and location together.

Introduction (Times New Roman 12, Bold, Aligned Left, Single Line, 0pt before 6pt after)

Understanding how animals behave around humans is an important part of studying urban wildlife. In places such as Central Park, squirrels regularly interact with people in different ways. Some squirrels approach humans, some avoid them, while others show neutral behaviour. These interactions may be influenced by several factors, including squirrel activity, environmental conditions, time of observation, and geographic location within the park. Analysing these patterns can help provide insight into how wildlife adapts to busy urban environments.

The research question explored in this project is: *Can we predict whether a squirrel will approach or avoid humans based on its behaviour, location, and time of observation, and which features are most influential?* In other words, the purpose of this report is to investigate whether squirrel and human interaction behaviour can be predicted using behavioural, spatial, temporal, and environmental features.

The analysis used data from the 2018 Central Park Squirrel Census dataset, which contains information about squirrel observations in Central Park, New York City. It should be noted that the dataset includes behavioural information such as running, climbing, foraging, and vocalisation, along with environmental and location-based variables. Also, other environmental information was also merged into the dataset to provide wider context for the analysis.

To answer the research question, several data preprocessing techniques were applied, including feature engineering, missing value handling, data integration, and spatial grouping. Correlation analysis, supervised machine learning models, and clustering techniques were then used to explore patterns and relationships within the data.

Methodology

2.1 Data preprocessing

The research question of this study is to predict whether a squirrel will approach or avoid humans based on its behaviour, location, and time of observation. However, there are columns that indicate "Approaches", "Runs from", and "Indifferent", and these are the answers that we are trying to predict, since these variables represent squirrel behaviour toward human outcomes, including them as input features would create target leakage and prevent meaningful prediction. Therefore, including these three columns as features makes the model simply predict the target variable itself, effectively allowing it to 'cheat' by using the answer to predict the answer. Thus, we transformed these three columns into one **target variable**, human_reaction: 0 = Runs from = True, 1 = Indifference (Other), 2 = Approaches = True.

In this case, each behavioural column was presented as a True/False column but having too many binary columns makes it difficult to find meaningful patterns. So, **feature engineering** was applied and was grouped into four scores: *active_score*,*foraging_score*, *vocal_score*, and *alert_score* . This reduces dimensionality and makes it easier to identify patterns such as whether highly active squirrels tend to avoid humans. Time of day (AM/PM) was encoded as 0 and 1, and dates were converted into weekday numbers (0–6). As an alternative, *pd.get_dummies()* can be used. However, binary encoding is more appropriate because it only has 2 variables (AM and PM) which possibly generate duplication of information. These variables were included because people normally prefer visiting the park on weekends and in the afternoon, which may directly influence our research question.

Data Integration was used to obtain location-based information. Zone-level features from *hectare.csv* — crowd conditions (*conditions_enc*), presence of threat animals (*has_threat*), and total squirrels observed (*total_squirrels*) — were merged using a left join to preserve all squirrel records. It is essential since the research question includes location specifically. Furthermore, inconsistent entries such as combined states (e.g., *Calm, Busy*) and a typo (*Medium*) were standardized to *Moderate* before encoding in order to give accurate values for analysing how zone level affects squirrel behaviour.

Handling missing Values takes a significant part in this stage since it is used to avoid errors and biased prediction due to incomplete data. Categorical variables with low missingness (*Age*, *Primary Fur Color*) were replaced using mode. *Highlight Fur Color* was removed because of 36% missingness and directly related to the research question. After merging, NaN values in *has_threat* and *total_squirrels* means that there was no observation at that time in the area, so it is logical to fill it with 0.

It is difficult to interpret numbers such as coordinates and squirrel numbers with our research question. **Discretization** was conducted to change values into categories which brings clearer analysis of whether density and location affect squirrels' behavior. The X, Y coordinates were transformed into four geographic regions (NE/NW/SE/SW) based on the median, and as a result, four regions had similar sizes. Dividing into two (North/South) is too simple to miss specific patterns depending on the location. Conversely, if you subdivide too much, the number of samples in each area decreases, which leads to being less reliable. Therefore, it is a suitable choice that can manage interpretability and sample size by dividing the area into four to meet all these conditions. Moreover, squirrel densities were classified as low, medium, and high using the IQR bound (Q1 = 4.5, Q3 = 12.0). This brings us an improved interpretation of locations in relation to squirrels' behaviour toward humans.

Also, **outlier detection** was performed for squirrel density and spatial coordinates using the IQR method. Instead of removing unusual observations, they were kept because areas with high squirrel numbers may still contain useful information about squirrel and human interaction behaviour.

Overall, the preprocessing part transformed the raw squirrel observations into a more structured and interpretable dataset suitable for future stages.

2.2 Correlation

Correlation analysis was used to explore the relationship between squirrel and human interaction behaviour and different behavioural, environmental, spatial, and temporal features in the dataset. It should be noted that, the main purpose of this step was to understand which features were more strongly related to squirrel behaviour toward humans before building supervised learning models. Also, this analysis helped identify whether there were any noticeable patterns between squirrel activities and human interaction categories.

The target variable used in this part was **human_reaction**, which showed three different squirrel behaviours toward humans: avoiding, being neutral, and approaching. Several variables were selected for the correlation analysis because they were directly related to the research question, which included the engineered behavioural scores such as active_score, individual behaviours such as running, and spatial coordinates (X and Y), weekday information, shift information, squirrel density, environmental conditions, and threat presence.

Some variables such as Identifier columns and text-heavy variables were removed because they do not provide meaningful numerical relationships. Also, some categorical variables were excluded because they were not suitable for direct Pearson correlation calculations without additional transformation.

Pearson correlation was selected because it provides a simple and easy to understand measurement of the linear relationship between two numerical variables. Correlation values range between -1 and 1. A value closer to **1** represents a stronger positive relationship, which means both variables tend to increase together. While, a value closer to **-1** represents a stronger negative relationship, meaning that one variable tends to decrease when the other increases. Values close to **0** show weak or no evident linear relationship.

The correlation results were visualised using a feature correlation bar chart which showed how strongly each feature was associated with squirrel and human interaction behaviour. Also, a separate distribution bar chart was created to show the number of observations in each human interaction feature.

Overall, Pearson correlation was chosen because it allowed **quick comparison** between many numerical variables and provided a clear overview of feature relationships. However, this method also has limitations. Many variables in the dataset were binary or categorical, which means Pearson correlation may not fully capture more complex or non-linear relationships.

2.3 Supervised Learning

Using several supervised learning models, we attempted to predict squirrel reactions toward humans and reactions towards humans based on behavioural, spatial, temporal and environmental features. The target variable, human_reaction, consisted of three discrete behavioural categories, being: Runs Away (0), Indifferent (1) and Approaches (2). A supervised learning model was appropriate for this task as historical data regarding behavioural outcomes were available for model training and evaluation. The purpose of this prediction task is to identify whether squirrel behaviours and environmental factors could meaningfully explain the behaviour of squirrel and human interactions. Understanding the way squirrels interact with humans could assist in explaining how urban wildlife is influenced by exposure to humans.

Predictor feature variables were chosen due to their likelihood to impact and influence squirrel reactions to humans. The chosen predictor variables were engineered behavioural scores (active_score, foraging_score, vocal_score, alert_score); time features (shift_enc, weekday); area features (zone, density_level, conditions_enc, has_threat, outlier_variable); spatial features (X, Y); and squirrel characteristics (age, primary fur colour, location). The behavioural scores were engineered by taking related binary datapoints and aggregating them into compounded scores. Due to the reduced dimensionality, broader behavioural patterns are easier to identify and analyse as opposed to taking each individual datapoint and trying to observe patterns within them. Also, this reduced sparsity and improved interpretability during clustering and supervised learning. Time features were chosen in order to identify differences in squirrel behaviour over time and different time periods. Environmental and spatial features were chosen in order to capture the surrounding ecological and habitat factors that could influence the behaviour of the squirrels. All of these factors cumulatively were chosen as they were expected to influence a squirrel's mannerisms and behaviour in urban environments.

Light amounts of data preprocessing were conducted before the supervised learning models were created in order to account for data quality, reducing inconsistencies and to ensure that the models could perform to a higher quality. Missing values from categorical features (zone, density_level, age, primary fur colour, location) were imputed using the mean, while missing values from the numerical features (active_score, foraging_score, vocal_score, alert_score, shift_enc, weekday, conditions_enc, has_threat, outlier_variable, X, Y) were imputed using the mean. Categorical variables were one-hot encoded to convert non-numerical categories into numerical representations suitable for machine learning models. Numerical variables were standardised using feature scaling in order to improve the performance of logistical regression, which can be sensitive to issues surrounding feature magnitudes. The dataset was then split into training and testing subsets using an 80/20 split, with 80% of data being used for training and 20% for testing.

Logistic regression was chosen as the first supervised learning model mainly for its suitability for multiclass classification problems, as well as its interpretability and efficiency. The model was used to predict squirrel reactions and interactions with humans based on the features previously described. Logistic Regression was also suitable for this task, due to the way it estimates class probabilities using weighted relationships between predictor variables and the potential outcomes.

A maximum iteration value of 2000 was chosen for the model in order to ensure convergence during training. Prediction bias was reduced by applying balanced class weights, as otherwise bias could have occurred towards the "Indifferent" class as it covered majority of the dataset. As previously mentioned, a 80/20 stratified train-test ratio was used for the data while numerical variables were standardised and categorical variables were one-hot encoded. A 5-fold cross validation was conducted after the fact in order to evaluate model performance. However, Logistic Regression assumes approximately linear relationships between features and target classes, which could limit its ability to capture more complex behavioural patterns.

Random Forest was chosen as the second supervised learning model to be used for this task. This was mainly due to its ability to identify nonlinear relationships between the predicting factors. These predicting factors may contain patterns that are nonlinear and difficult to separate otherwise, in terms of understanding context from environmental data, and such, Random Forest was decided to be a quality fit for the task.

Unlike Logistic Regression, Random Forest does not necessarily require feature scaling, however, the same preprocessing tasks previously mentioned in 2.3.3 and 2.3.4 were implemented for the Random Forest model as well in order to ensure fair model comparison after the fact.

The Random Forest model used was configured using 200 decision trees, a maximum tree depth of 12 and a minimum split size of 5. Once again, balanced class weights were chosen in order to reduce bias towards the dominant class, "Indifferent". Evaluation for the Random Forest model was conducted using 5-fold cross validation as well.

It was made sure that both models were evaluated fairly, and evaluated using the same methodology and metrics. Using consistent evaluation methods ensured meaningful comparison between the models. The evaluation metrics used were accuracy, precision, recall and F1-score. Accuracy measures the overall percentage of correct predictions, while precision and recall measure the reliability and sensitivity of predictions relating to each behavioural class. It should be noted that, accuracy alone was insufficient because the dataset was not balanced, with neutral behaviour dominating the target distribution. A weighted F1-score was prioritised as it balances precision and recall while taking into account the class imbalance within the dataset. A classification report was made to outline minority-class performance independently of the entire class frequency. Confusion matrices were used to identify and assess model bias towards a dominant class. On top of this, 5-fold cross validation was implemented again at the end. It should be noted that, Cross-validation was used to evaluate how consistently the models generalised across different subsets of the dataset.

2.4 Clustering

To provide additional unsupervised insights for the research question, two K-Means clustering analyses were conducted. The first clustering analysis focused on squirrel behavioural patterns, while the second focused on environmental and spatial conditions.

The research question aims to determine whether squirrels' approach or avoid humans based on behavioural and environmental factors. These factors include behaviours and locations. Therefore, clustering these two dimensions separately can examine whether distinct natural groups exist and how they relate to human interactions. Moreover, the clustering results were used to examine whether naturally occurring squirrel groups displayed different interaction patterns with humans.

K-Means Clustering was used because the selected behavioural and environmental variables were numerical and the method produces easily interpretable cluster centres for comparison. It also allowed the use of feature mean values to represent each cluster. In addition, the numeric values make the relationship between clusters and the human_reaction column more readable.

On the other hand, Hierarchical Clustering was considered but was eventually not used. This is because **Hierarchical Clustering** mainly identifies nested relationships between observations. However, the research question focuses more on identifying different reactions within each cluster rather than finding hierarchical relationships. Therefore, Hierarchical Clustering was not used in this process.

After selecting the clustering method, it was necessary to choose features that satisfied the research requirements. For behavioural clustering, behavioural features were selected. These features consisted of four different scores including activity, foraging, vocal, and alert scores. It should be noted that using engineered behavioural scores reduced feature redundancy and simplified the identification of broader behavioural patterns during clustering. Similarly, environmental features were selected for environmental clustering for the same reason.

There was also a data cleaning process before clustering. Missing values remaining after preprocessing were removed before clustering because distance-based clustering methods are sensitive to incomplete data. These missing values could strongly affect distance calculations, so removing them was necessary.

The next stage was normalisation, which adjusted the range of features using different scaling methods. This step rescales features to a similar numerical range. Without this process, features with larger numerical ranges would have more influence on distance calculations than other features. In this part, **MinMaxScaler** was used for behavioural clustering, while **StandardScaler** was applied to environmental clustering. Different scaling methods were selected because the behavioural and environmental features had different data distributions and value ranges. The ranges of active_score, foraging_score, vocal_score, and alert_score was quite similar since they all started from 0 and ended at values smaller than 5. Therefore, MinMaxScaler was suitable for behavioural clustering. However, environmental features had very different distributions. For example, coordinate variables X and Y are large decimal values, while has_threat is binary. Therefore, StandardScaler was applied to environmental clustering to rescale these substantially different variables. Importantly, scaling was necessary because K-Means clustering relies heavily on distance calculations, meaning features with larger numerical ranges could otherwise dominate the clustering process.

The Elbow Method was used to determine the optimal number of clusters, k, for the K-Means clustering analyses. This method examines the relationship between the number of clusters and the Sum of Squared Errors (SSE), which measures how closely data points are grouped around their cluster centres. As the number of clusters increases, the SSE value decreases because the data points become more tightly grouped. However, after a certain point, the improvement begins to slow down. This point, known as the "**elbow**", shows a suitable balance between cluster quality and model complexity.

Results Exploration and Analysis

3.1 Data Preprocessing

Several important patterns were identified during the preprocessing stage. The *human_reaction* variable showed a class imbalance that was biased to one side, 2,187 cases (72%), Runs from 658 cases (22%), and Approaches 178 cases (6%). As the research question is focused on predicting approach and runs from behavior, the lack of data in two classes limits the model's ability to learn meaningful patterns. This imbalance later influenced supervised learning performance.

In terms of location data, all coordinates are surprisingly inside the Central Park range, indicating that location-based features such as *zone* and *density_level* are accurately recorded. Moreover, 74 areas exceeding the upper limit of IQR standard (Q1=4.5, Q3=12.0) of 23.25 were detected, which indicates unusually high squirrel concentrations within specific park areas. These results were not removed, these observations were retained and flagged using the **outlier_variable** feature rather than removed from the dataset.

In the zone distribution, the results were concentrated in North East (1,328 cases) and South West (1,058 cases), while South East (633) was relatively small and North West only had four cases. This implies that squirrel habitats in the park are not evenly distributed, especially as the NorthWest zone contained a very small number of observations, meaning results from this region should be interpreted cautiously. It should be noted that, this unequal geographic distribution could mean that squirrel activity and observations may vary across different areas of Central Park. On the other hand, the density distribution was evenly distributed into Medium (1,862), Low (818), and High (343 cases), which provided sufficient variability to analyse differences in squirrel behaviour across different density conditions.

3.2 Correlation

The correlation analysis was conducted to examine the relationship between squirrel and human interaction behaviour and selected behavioural, spatial, temporal, and environmental variables.
The bar chart (figure 1) shows that **neutral behaviour** was the most common interaction type in the dataset, while approach behaviour happened much less frequently. This shows that the dataset was imbalanced, with a much larger number of neutral observations compared to the minority interaction categories.

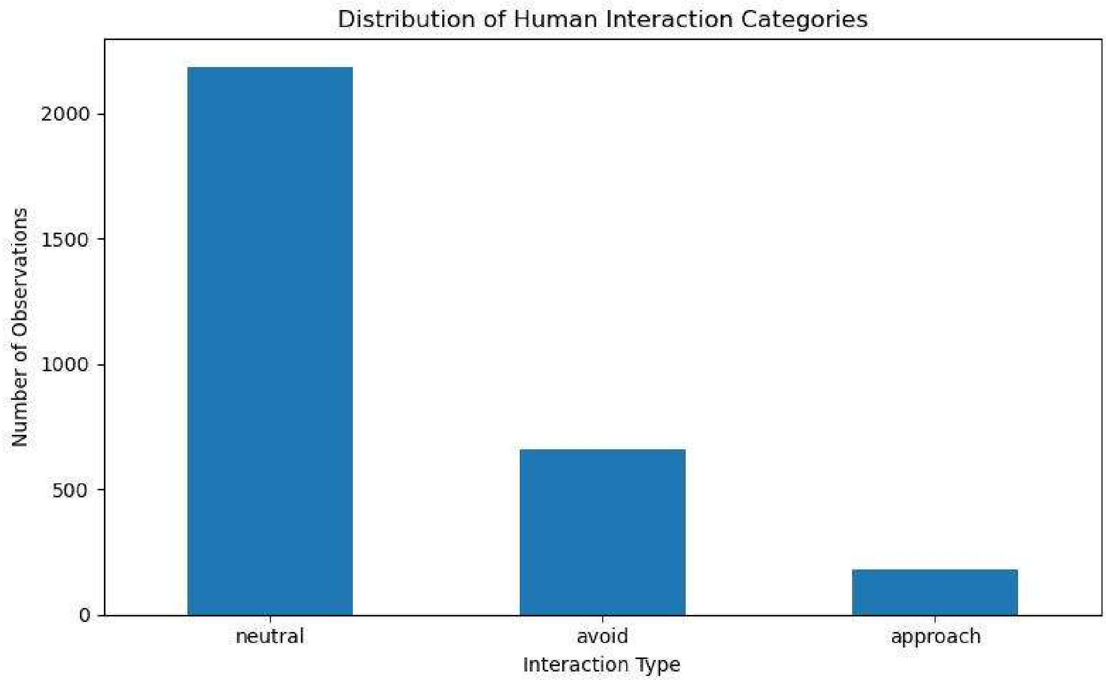

Figure 1 distribution of human interaction categories bar chart

Pearson correlation analysis was then used to measure the relationship between the selected features and the **human_reaction** target variable. The correlation values are shown in Figure 2.

The correlation results showed that most variables had relatively weak relationships with squirrel and human interaction behaviour. None of the features produced a very strong positive or negative correlation, suggesting that squirrel and human interaction behaviour is not influenced by a single dominant factor.

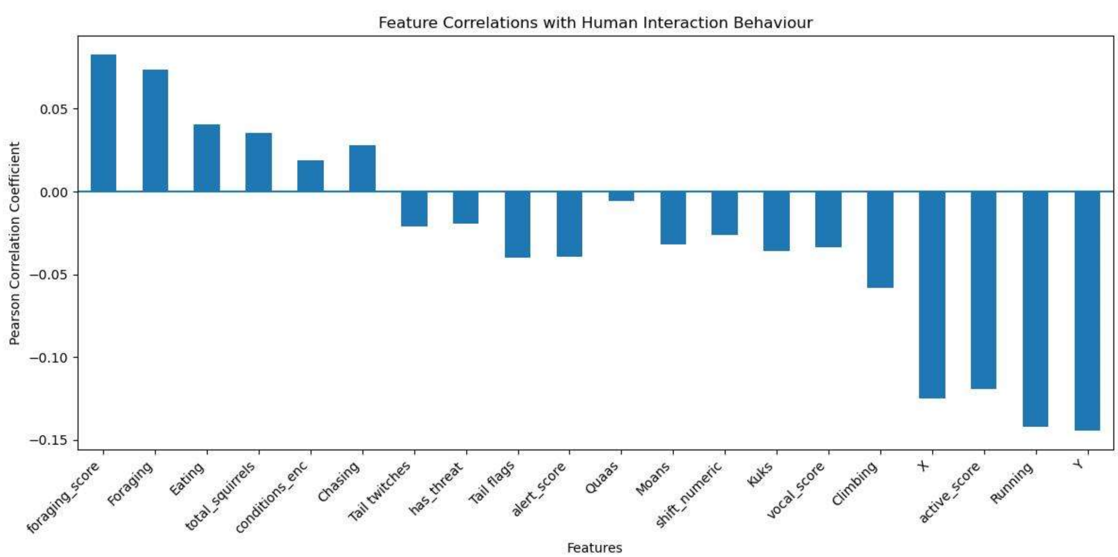

Figure 2 feature correlations with human interaction behaviour bar chart

Among the behavioural variables, **foraging_score** and **Foraging** showed the strongest positive relationships with human interaction behaviour. This suggests that squirrels engaged in food-seeking activities were more likely to display positive or neutral interactions with humans. In contrast, **Running** and **active_score** showed negative relationships with the target variable, indicating that highly active or reactive squirrels were more likely to avoid human interaction.

In addition, the geographical variables **X** and **Y** also showed noticeable negative correlations with human interaction behaviour. This suggests that squirrel and human interaction patterns may vary across different areas of Central Park. Environmental variables such as total_squirrels, conditions_enc, and has_threat showed weak but observable relationships with the target variable, which shows that environmental context may also influence squirrel behaviour toward humans.

Overall, the correlation analysis showed that squirrel and human interaction behaviour is likely influenced by multiple behavioural, environmental, and spatial factors together rather than a single feature alone. The generally weak pairwise correlations also suggested that more complex relationships may exist within the dataset, supporting the use of supervised learning models to further analyse interaction behaviour.

3.3 Supervised Machine Learning Models

The two separate supervised learning models were evaluated between themselves in order to find which of the two could better predict a squirrel's behaviour in an interaction with a human, using the previously defined predictor variables as input. This evaluation process focuses on comparing both the behaviour and the performance of the Logistic Regression model and the Random Forest model in terms of answering the research question: whether squirrel and human interactions can be predicted by environmental and other factors.

The model comparison was conducted using accuracy, precision, recall and a weighted F1-score as well as confusion matrix analysis. This approach was considered and implemented due to the target variable containing imbalanced behavioural classes, which means that if only accuracy was used, it could provide misleading conclusions. Also, since neutral behaviour dominated the dataset, accuracy alone could produce misleading conclusions by rewarding models that mainly predicted the majority class.

The two selected supervised learning models were also chosen specifically for their contrasting strengths, with Logistic Regression providing very strong interpretability while Random Forest offering a better ability to model non-linear relationships. This evaluation therefore not only highlighted predictive performance, but also the model's behaviour, minority-class prediction and their broader importance to answering the research question.

Table 1: Comparison between Logistic Regression results and Random Forest results

Model	Accuracy	Precision	Recall	Weighted F1
Logistic Regression	0.428	0.658	0.428	0.480
Random Forest	0.736	0.663	0.736	0.667

Table 2: Classification Report for the Logistic Regression model

Class	Meaning	Precision	Recall	F1
0	Runs Away	0.32	0.51	0.39
1	Indifferent	0.81	0.40	0.53
2	Approaches	0.10	0.51	0.17

Table 3: Classification Report for the Random Forest model

Class	Meaning	Precision	Recall	F1
0	Runs Away	0.55	0.17	0.26
1	Indifferent	0.75	0.97	0.84
2	Approaches	0.00	0.00	0.00

Table 4: Random Forest model's feature importance

Feature	Importance
Y	0.221796
X	0.219111
weekday	0.079998
active_score	0.055957
foraging_score	0.048549
conditions_enc	0.043353
alert_score	0.034279
shift_enc	0.031510
has_threat	0.030243
density_level_Medium	0.022989
density_level_Low	0.021559
vocal_score	0.020064
Primary Fur Color_Cinnamon	0.019696
Primary Fur Color_Gray	0.018993
Location_Ground Plane	0.018656
Location_Above Ground	0.017937
Age_Adult	0.017165
Age_Juvenile	0.016208
density_level_High	0.014313
zone_NorthEast	0.012196
zone_SouthEast	0.010820
zone_SouthWest	0.010503
Primary Fur Color_Black	0.008751
outlier_variable	0.003823
zone_NorthWest	0.001503
Age_?	0.000027

Table 5: Logistic Regression Coefficients

Variable	Runs Away (0)	Indifferent (1)	Approaches (2)
zone_NorthEast	-0.177907	0.197772	-0.019866
zone_NorthWest	0.484912	-0.165365	-0.319547
zone_SouthEast	-0.066035	0.139222	-0.073186
zone_SouthWest	-0.214875	-0.129517	0.344391
density_level_High	-0.109590	0.001220	0.108370
density_level_Low	0.100281	0.034648	-0.134930
density_level_Medium	0.035405	0.006243	-0.041648
Age_?	-0.137360	0.203979	-0.066619
Age_Adult	0.119028	-0.051954	-0.067074
Age_Juvenile	0.044428	-0.109913	0.065485
Primary Fur Color_Black	0.140638	0.113286	-0.253924
Primary Fur Color_Cinnamon	-0.141078	-0.192795	0.333873
Primary Fur Color_Gray	0.026536	0.121621	-0.148157
Location_Above Ground	0.032421	0.238080	-0.270501
Location_Ground Plane	-0.006325	-0.195968	0.202293
active_score	0.170440	-0.092484	-0.077956
foraging_score	-0.079029	-0.061845	0.140874
vocal_score	0.087603	0.037615	-0.125217
alert_score	0.032803	-0.104864	0.072061
shift_enc	0.061200	0.003963	-0.065163
weekday	-0.021056	0.021166	-0.000110
conditions_enc	0.012384	0.017252	-0.029636
has_threat	0.081326	0.079855	-0.161181
outlier_variable	0.043859	0.053879	-0.097738
X	-0.127549	-0.046948	0.174497
Y	0.362583	-0.188897	-0.173686

As seen in Table 1, the Random Forest model achieved substantially better results than the Logistic Regression model, with an accuracy of 73.6% and a weighted F1-score of 0.667, compared to Logistic Regression, which had an accuracy of 42.8% and a weighted F1-score of 0.480. However, when analysing the classification reports in Tables 2 and 3, it can be seen that both models were affected by class imbalance, with the Random Forest model heavily favouring the "Indifferent" (1) class with a recall of 0.97, while the Logistic Regression model produced more balanced predictions across all behavioural classes. Random Forest also failed to correctly classify any of the "Approaches" data, which is a large issue for the supervised learning model. In contrast, the Logistic Regression model produced far more balanced predictions across all classes, albeit at the sacrifice of accuracy.

The two different supervised learning models provided quite different predictive behaviours and dependencies on different input features. As previously mentioned, the Logistic Regression model produced more balanced predictions across the different input features, while the Random Forest model highly favoured the dominant “Indifferent” class. This shows that the Random Forest model focused more strongly on dominant behavioural patterns, while the Logistic Regression model was able to better identify smaller patterns and behavioural identities.

3.4 Clustering

Two K-Means clustering analyses were conducted to explore hidden behavioural and environmental patterns related to squirrel and human interaction behaviour. The first clustering focused on squirrel behavioural patterns, while the second focused on environmental and spatial conditions.

Based on the Elbow Method, k = 5 was selected for behavioural clustering and k = 3 for environmental clustering because these values produced meaningful and interpretable cluster structures.

This heatmap (Figure 3) presents the mean values of behavioural features consisting of four different scores. The colour bar on the right represents the magnitude of the values. It shows that each cluster has its own characteristics. Cluster 0 had a high value of foraging_score, so it was defined as “Forager”. Cluster 1, with a high active_score, was considered “Active”. In Cluster 2, both foraging_score and alert_score was higher than 1, so it was renamed “Alert Forager”. Cluster 3 showed the strongest food-seeking behaviour and was therefore labelled “Most Forager”. Cluster 4 was defined as “Alert Active” because of its high active_score and alert_score.

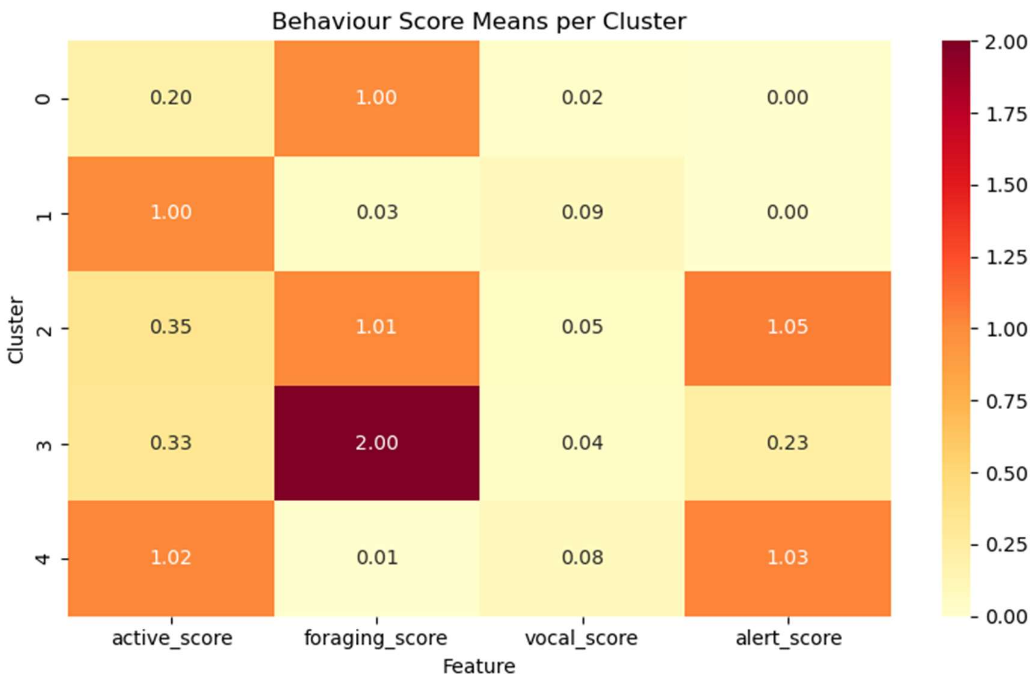

Figure 3 Behaviour Score Means per Cluster

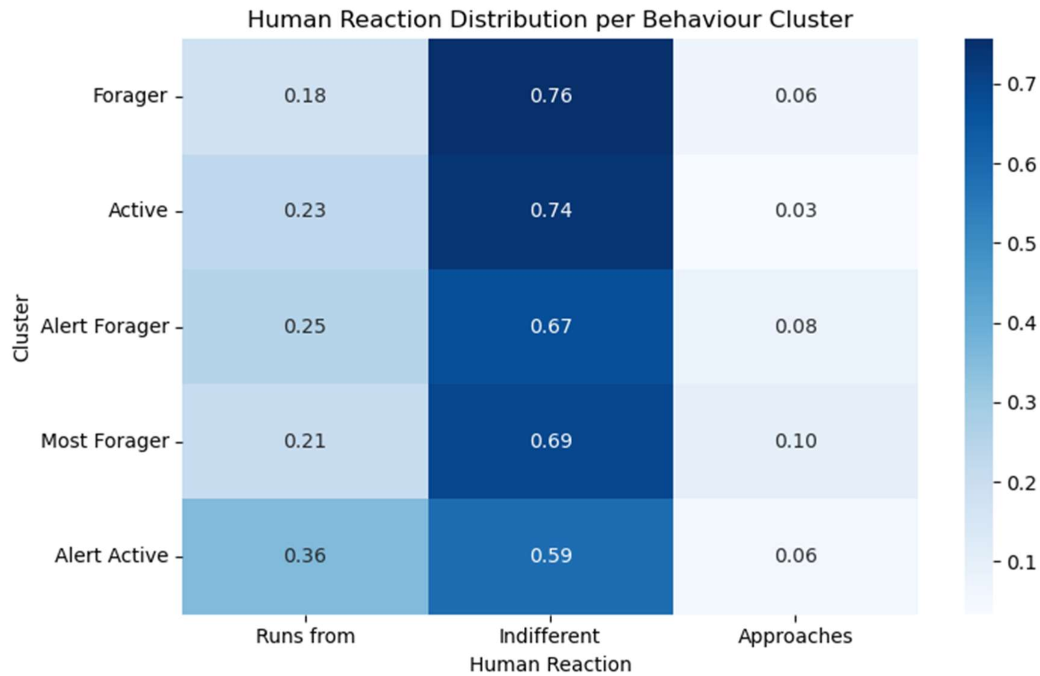

Figure 4 Human Reaction Distribution per Behaviour Cluster

As shown in Figure 4, indifferent behaviour was the most common reaction in each cluster. This can be supported by the percentage of "Indifferent" observations in each cluster, all of which were higher than 50%. Another finding was that squirrels with higher alert_score values seemed more likely to avoid humans. The "Alert Forager" and "Alert Active" clusters had relatively higher rates of "Runs from" behaviour than other clusters. This suggests that squirrels with higher alert behaviour may be more cautious around humans. Additionally, the "Most Forager" cluster showed the highest proportion of "Approaches" behaviour, suggesting that food-seeking squirrels may be more tolerant of nearby humans. This finding was consistent with the earlier correlation analysis, where foraging-related behaviours showed positive relationships with squirrel and human interaction behaviour.

The heatmap (Figure 5) represents three clusters using the mean values of environmental features. For conditions_enc, Cluster 0 and Cluster 2 had relatively higher values of 1.27 and 1.39. In terms of threat level, Cluster 0 showed the highest threat presence while also containing relatively low squirrel density, whereas Cluster 2 showed the lowest threat level. Moreover, the highest value of total_squirrels belonged to Cluster 2. Based on these findings, Cluster 0 was labelled "High-Threat Low-Density", Cluster 1 was labelled "Medium Environment", and Cluster 2 was labelled "High-Density Busy".

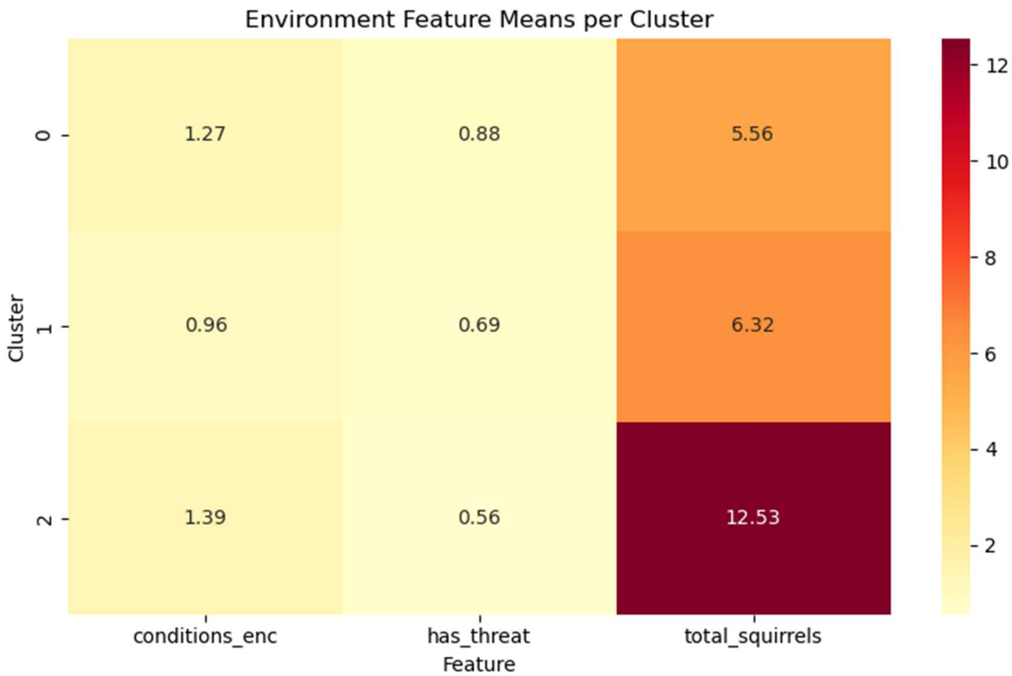

Figure 5 Environment Feature Means per Cluster

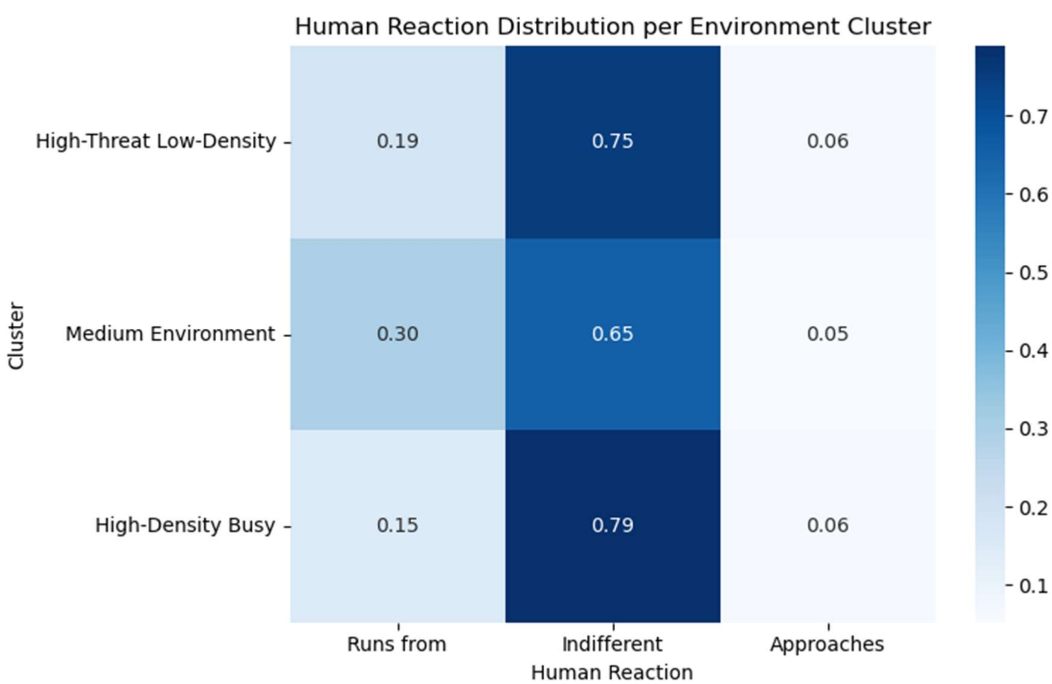

Figure 6 Human Reaction Distribution per Environment Cluster

As shown in Figure 6, the most common squirrel reaction was "Indifferent", meaning that squirrels most commonly displayed neutral behaviour toward humans. The proportions of indifferent behaviour in the three clusters were 0.75, 0.65, and 0.79. Furthermore, an interesting finding was that squirrels in the "Medium Environment" cluster were more likely to avoid humans than squirrels in other clusters. The "Runs from" rate in this cluster was 0.30, which was relatively higher than the other clusters at 0.19 and 0.15. In addition, approach behaviour remained relatively uncommon across all clusters, with proportions of 0.06, 0.05, and 0.06 respectively. The environmental clusters suggest that squirrel populations experience noticeably different ecological conditions across the park.

These findings were also consistent with the supervised learning results. Both clustering and supervised learning pointed to the same conclusion that "Indifferent" was the most common squirrel reaction toward humans. Moreover, clustering results showed that indifferent behaviour had the highest proportion across all behavioural and environmental clusters, suggesting that this pattern remained consistent regardless of environmental conditions.

Returning to the research question, the clustering analysis suggests that squirrel-human interaction behaviour is partially influenced by both behavioural habits and environmental conditions. For example, alert squirrels showed a stronger tendency to avoid humans, while foraging squirrels showed a slightly higher tendency to approach humans. This suggests that internal behavioural states may influence squirrel-human interactions.

Additionally, both clustering and supervised learning showed that it was difficult to predict minority classes, especially "Approaches". This was likely because very few "Approaches" observations existed in the dataset, making it difficult for both methods to identify stable patterns.

Overall, the clustering analysis suggests that squirrel and human interaction behaviour is influenced by both behavioural tendencies and environmental context. Highly active squirrels were more likely to avoid humans, while strongly foraging squirrels showed slightly higher approach behaviour. The clustering results also reinforced the supervised learning and correlation findings by showing that squirrel-human interaction behaviour is complex, spatially dependent, and influenced by multiple interacting factors rather than a single dominant feature.

Discussion and Interpretation

- **Class Imbalance and Model Behaviour:** One important result was that most squirrels showed neutral behaviour toward humans. This imbalance affected the supervised learning models. Random Forest achieved higher overall accuracy, but it mainly predicted the dominant neutral class and could not correctly classify the "Approaches" class. Logistic Regression achieved lower accuracy, but it produced more balanced predictions across the classes.
- **Spatial and Environmental Influence**: Location was one of the strongest factors affecting squirrel and human interaction behaviour. The Random Forest model showed that the spatial variables X and Y were the most important predictors suggesting that squirrel behaviour changes across different areas of Central Park.
- **Behavioural Patterns and Feature Engineering**: The engineered behavioural scores were useful in correlation analysis, supervised learning, and clustering. The clustering analysis identified meaningful squirrel groups such as "Forager" and "Alert Active". Food-seeking behaviours showed more positive relationships with human interaction behaviour, while highly active and alert squirrels were more likely to avoid humans. Also, the "Most Forager" group showed the highest level of approach behaviour, meaning that squirrels searching for food could become more comfortable around humans.
- **Complex Relationships Between Features**: The correlation analysis showed weak individual relationships between most features and squirrel and human interaction behaviour. However, Random Forest performed much better than Logistic Regression. This could mean that squirrel and human interaction behaviour depends on more complex relationships between multiple behavioural and environmental factors rather than one single feature alone. In addition, the clustering analysis supported this finding by identifying different behavioural and environmental groups linked to different interaction patterns.

Limitations and Improvement Opportunities

The largest limitation of this project was the strong class imbalance within the dataset, especially the small number of "Approaches" observations. This reduced the models' ability to accurately learn minority-class behaviour. In addition, Pearson correlation mainly captures linear relationships and may not fully represent more complex behavioural interactions. Also, K-Means clustering assumes simple cluster structures, which may not perfectly represent real behavioural patterns.

Future work could improve the analysis by collecting larger and more balanced datasets and using more advanced machine learning techniques to better capture complex behavioural relationships.

Conclusion

This report was trying to answer if squirrel behaviour toward humans could be predicted using behavioural, environmental, spatial, and time-related features from the 2018 Central Park Squirrel Census dataset. To answer the research question, data preprocessing, correlation analysis, supervised learning, and clustering techniques were used to study squirrel and human interaction behaviour.

The results showed that squirrel and human behaviour is influenced by several factors together rather than one single feature. Correlation analysis showed weak individual relationships between most variables and interaction behaviour, while the supervised learning models revealed more complex patterns in the data. Random Forest performed better than Logistic Regression, showing that squirrel behaviour depends on more complex relationships between behavioural and environmental factors. The analysis also showed that location was one of the strongest influences on squirrels' behaviour.

Also, Clustering analysis identified different behavioural and environmental squirrel groups linked to different interaction patterns, which supports correlation and supervised learning results. Food-seeking squirrels showed slightly more approach behaviour, while highly active and alert squirrels were more likely to avoid humans. It should be emphasized that, neutral behaviour was the most common squirrel reaction toward humans.

In conclusion, this analysis provides useful results of how squirrels behave toward humans. The findings show that behaviour, environment, and location all play a role in squirrel and human interactions.

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